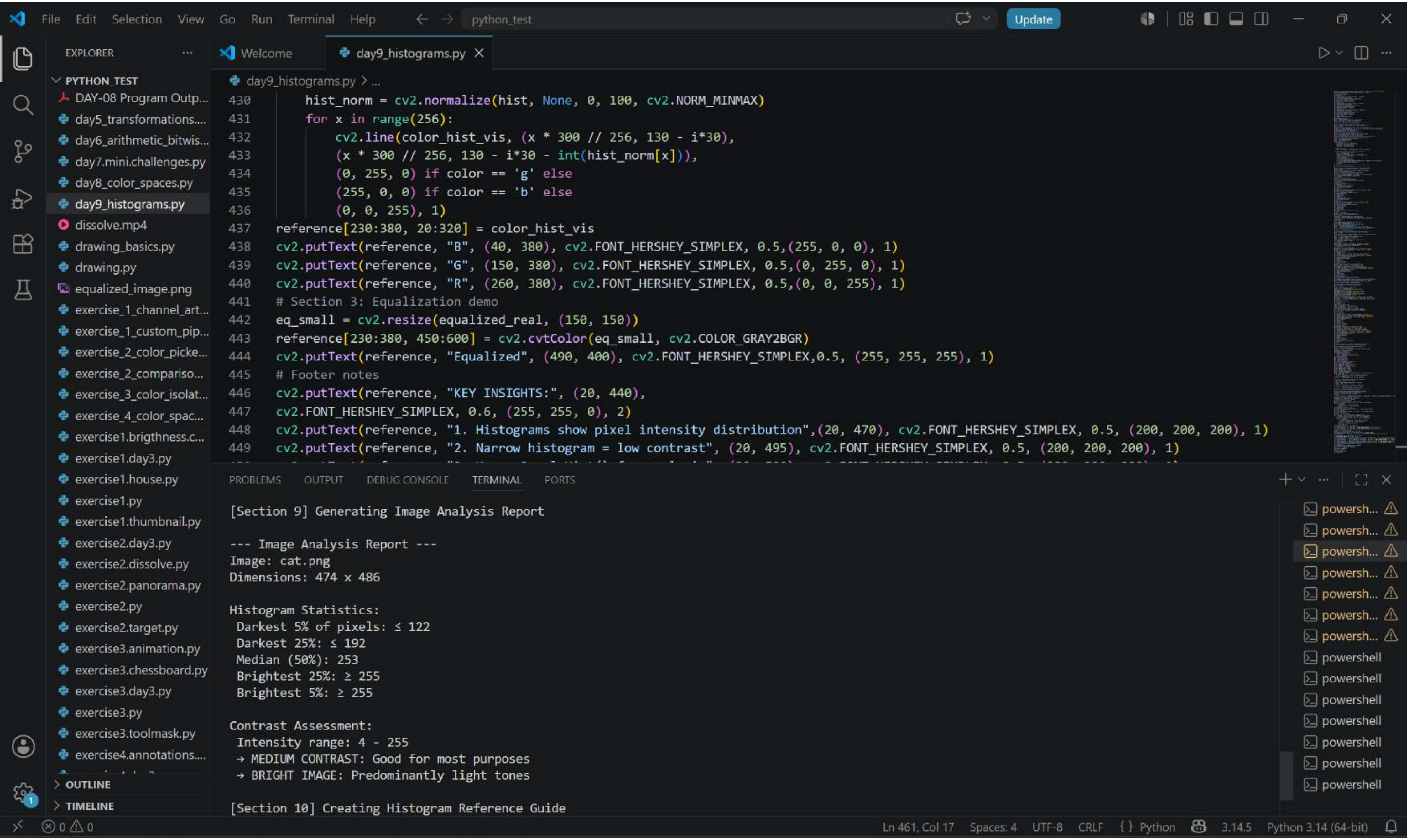

File Edit Selection View Go Run Terminal Help
python_test
Update
EXPLORER
PYTHON_TEST
DAY-08 Program Outp...
day5_transformations....
day6_arithmetic_bitwis...
day7.mini.challenges.py
day8_color_spaces.py
day9_histograms.py
dissolve.mp4
drawing_basics.py
drawing.py
equalized_image.png
exercise_1_channel_art...
exercise_1_custom_pip...
exercise_2_color_picke...
exercise_2_compariso...
exercise_3_color_isolat...
exercise_4_color_spac...
exercise1.brigthness.c...
exercise1.day3.py
exercise1.house.py
exercise1.py
exercise1.thumbnail.py
exercise2.day3.py
exercise2.dissolve.py
exercise2.panorama.py
exercise2.py
exercise2.target.py
exercise3.animation.py
exercise3.chessboard.py
exercise3.day3.py
exercise3.py
exercise3.toolmask.py
exercise4.annotations....
OUTLINE
TIMELINE
Welcome
day9_histograms.py
day9_histograms.py > ...
hist_norm = cv2.normalize(hist, None, 0, 100, cv2.NORM_MINMAX)
for x in range(256):
cv2.line(color_hist_vis, (x * 300 // 256, 130 - i*30),
(x * 300 // 256, 130 - i*30 - int(hist_norm[x])),
(0, 255, 0) if color == 'g' else
(255, 0, 0) if color == 'b' else
(0, 0, 255), 1)
reference[230:380, 20:320] = color_hist_vis
cv2.putText(reference, "B", (40, 380), cv2.FONT_HERSHEY_SIMPLEX, 0.5,(255, 0, 0), 1)
cv2.putText(reference, "G", (150, 380), cv2.FONT_HERSHEY_SIMPLEX, 0.5,(0, 255, 0), 1)
cv2.putText(reference, "R", (260, 380), cv2.FONT_HERSHEY_SIMPLEX, 0.5,(0, 0, 255), 1)
# Section 3: Equalization demo
eq_small = cv2.resize(equalized_real, (150, 150))
reference[230:380, 450:600] = cv2.cvtColor(eq_small, cv2.COLOR_GRAY2BGR)
cv2.putText(reference, "Equalized", (490, 400), cv2.FONT_HERSHEY_SIMPLEX,0.5, (255, 255, 255), 1)
# Footer notes
cv2.putText(reference, "KEY INSIGHTS:", (20, 440),
cv2.FONT_HERSHEY_SIMPLEX, 0.6, (255, 255, 0), 2)
cv2.putText(reference, "1. Histograms show pixel intensity distribution",(20, 470), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (200, 200, 200), 1)
cv2.putText(reference, "2. Narrow histogram = low contrast", (20, 495), cv2.FONT_HERSHEY_SIMPLEX, 0.5, (200, 200, 200), 1)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
[Section 9] Generating Image Analysis Report
--- Image Analysis Report ---
Image: cat.png
Dimensions: 474 x 486
Histogram Statistics:
Darkest 5% of pixels: ≤ 122
Darkest 25%: ≤ 192
Median (50%): 253
Brightest 25%: ≥ 255
Brightest 5%: ≥ 255
Contrast Assessment:
Intensity range: 4 - 255
→ MEDIUM CONTRAST: Good for most purposes
→ BRIGHT IMAGE: Predominantly light tones
[Section 10] Creating Histogram Reference Guide
powersh...
powershell
Ln 461, Col 17 Spaces: 4 UTF-8 CRLF { } Python 3.14.5 Python 3.14 (64-bit)

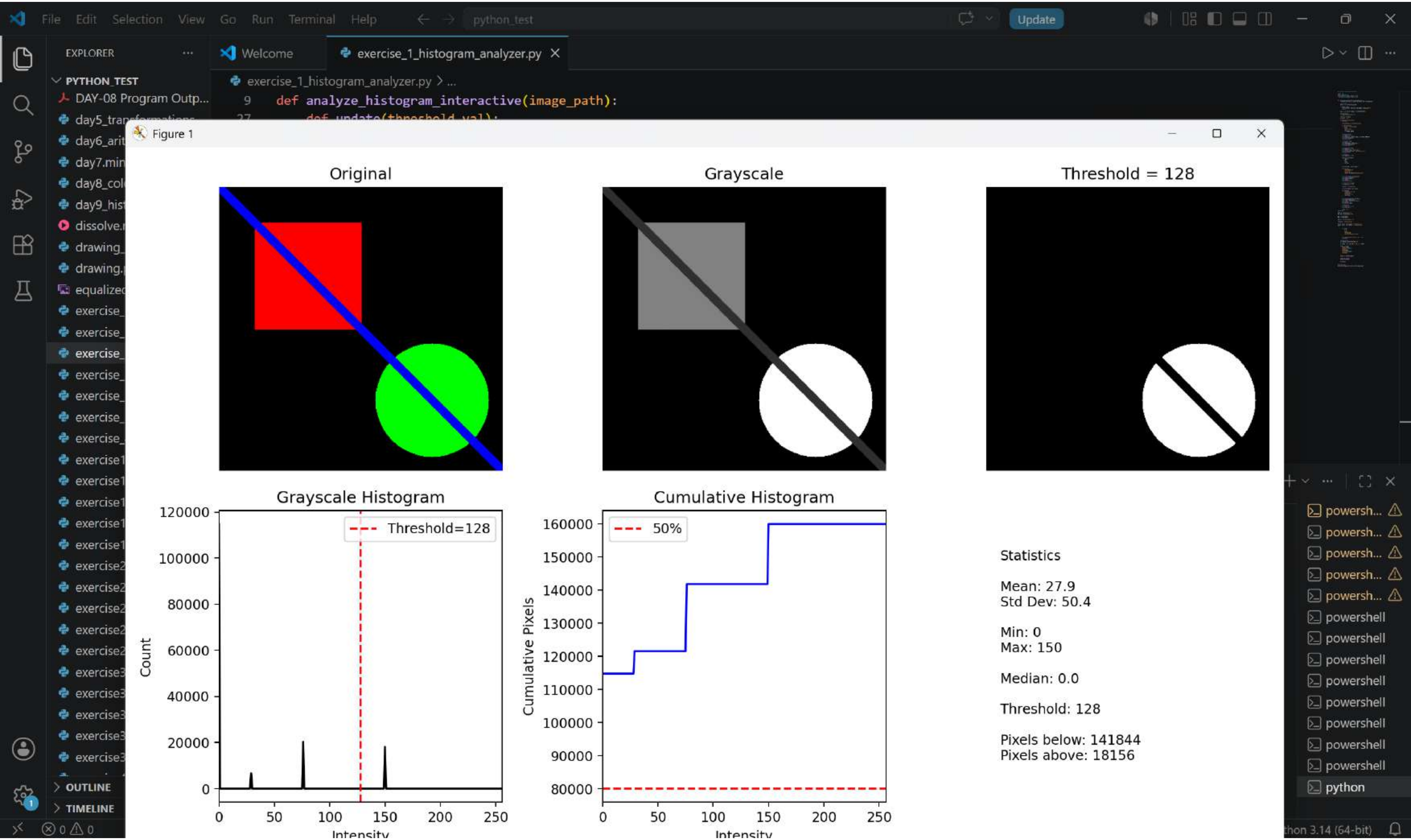
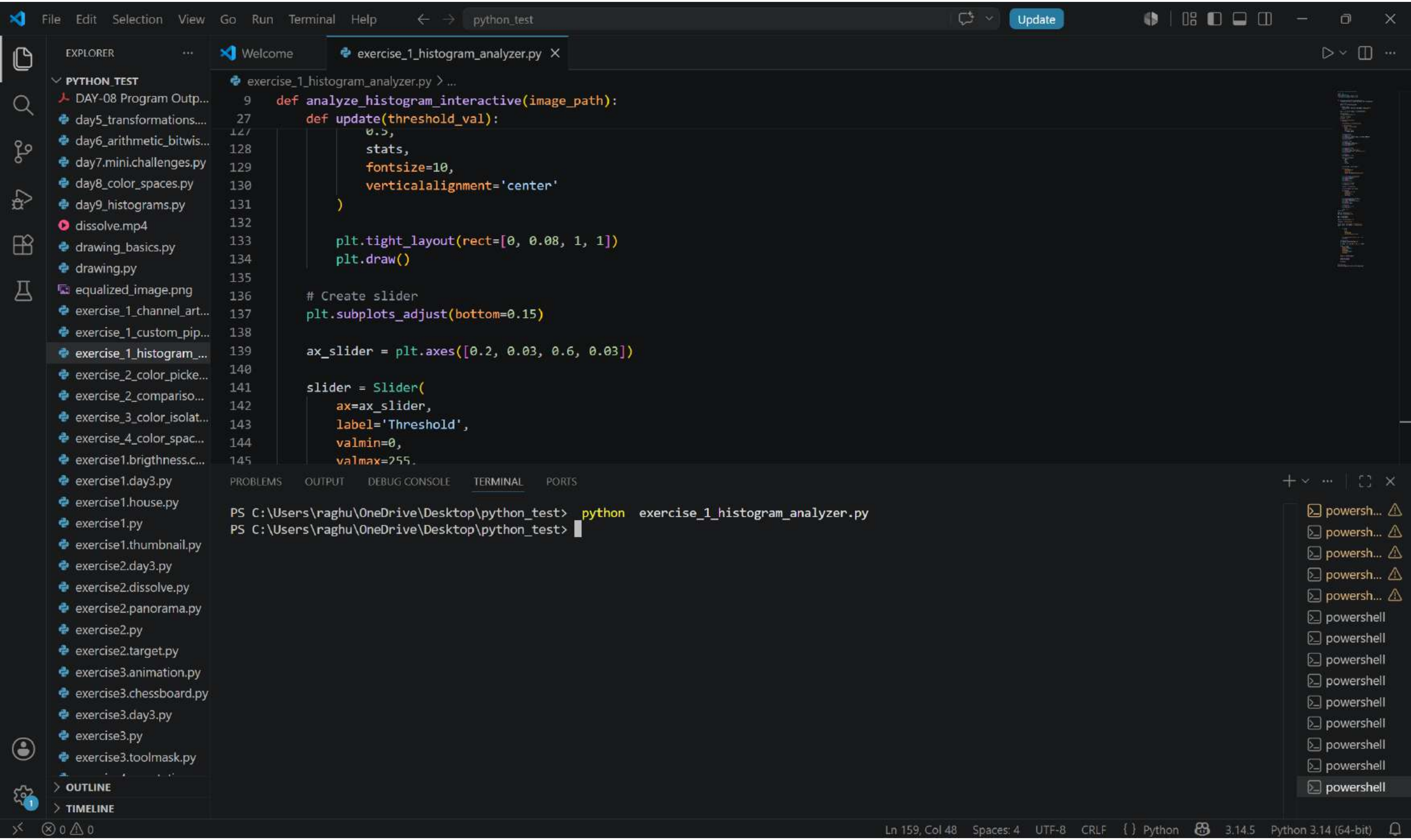
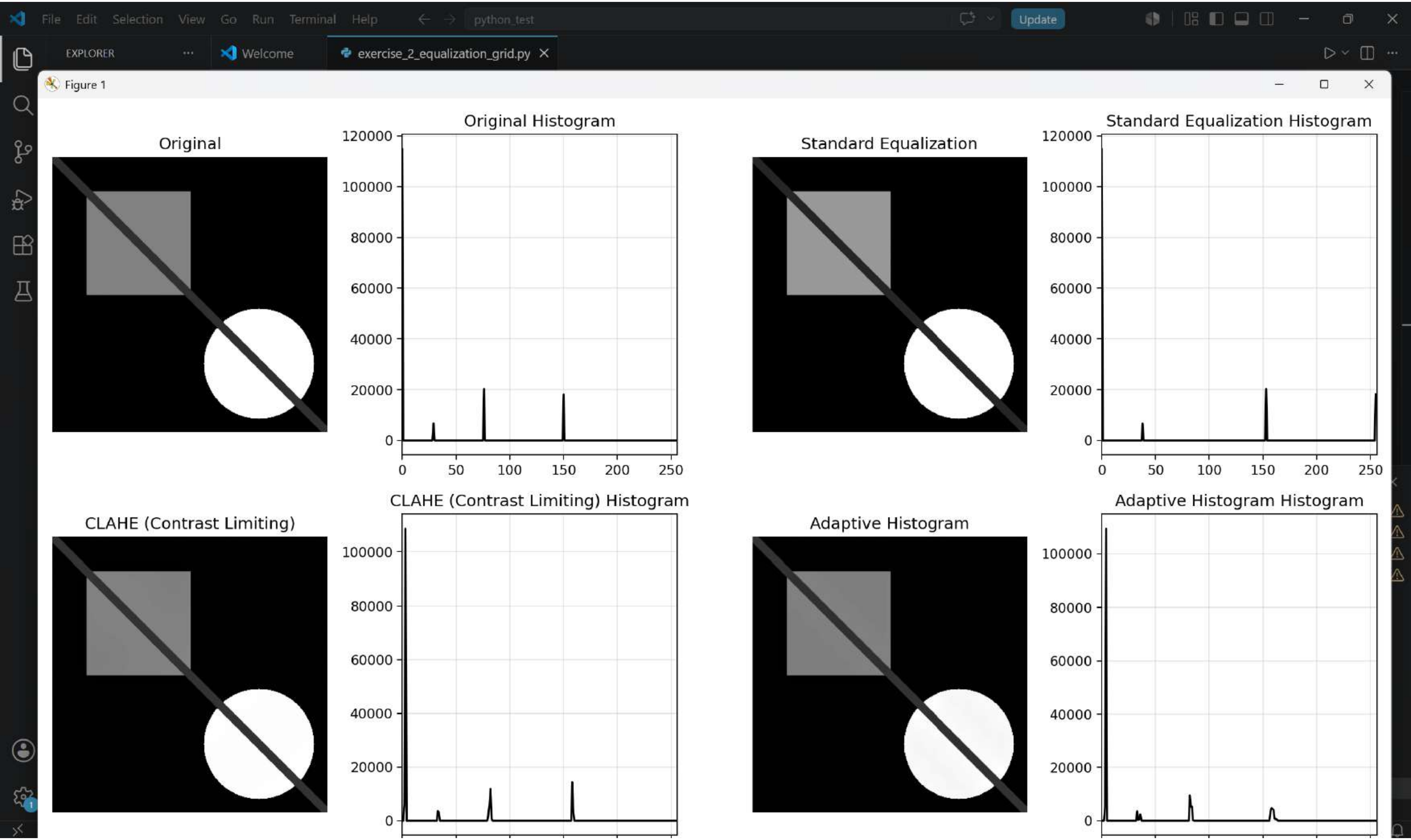
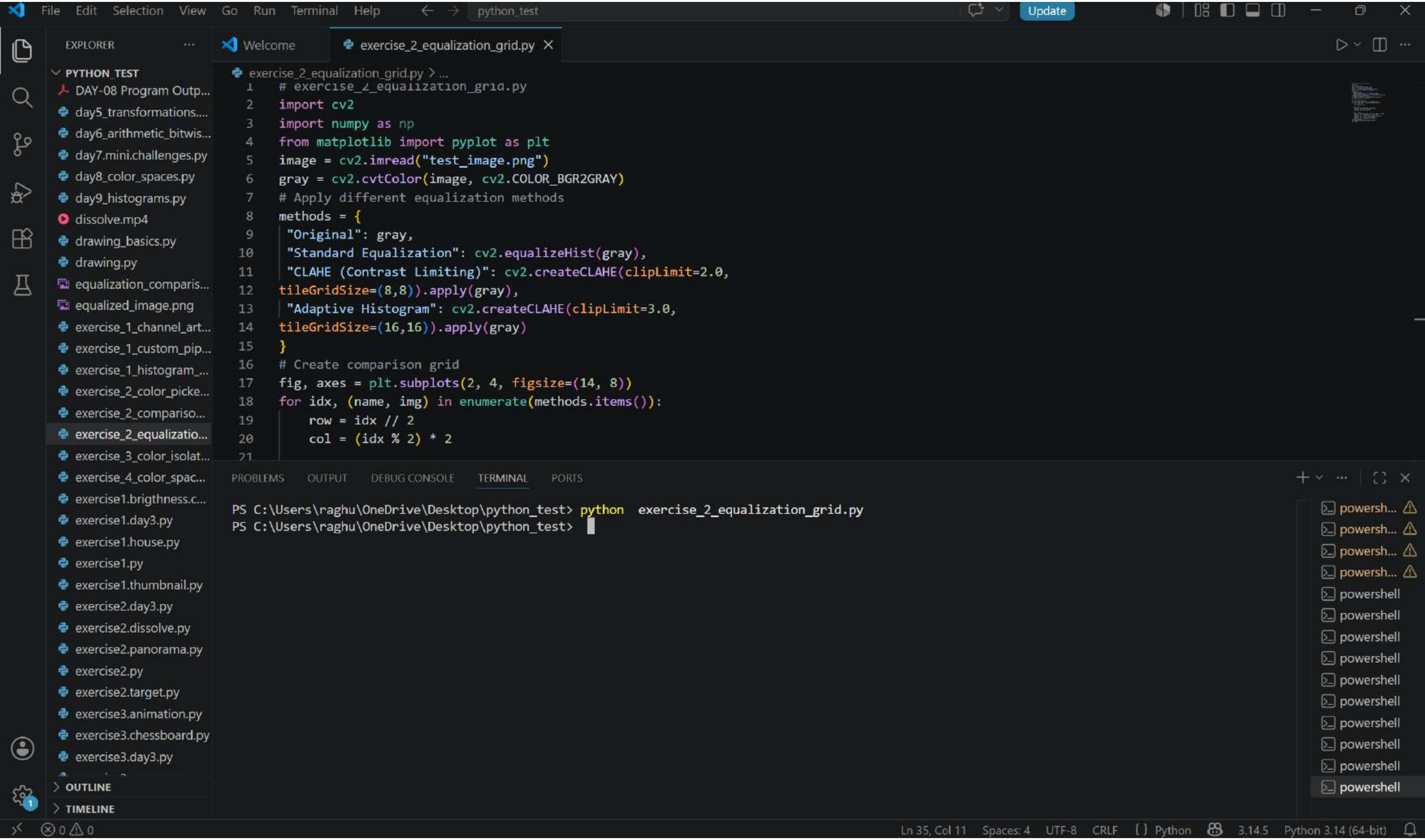


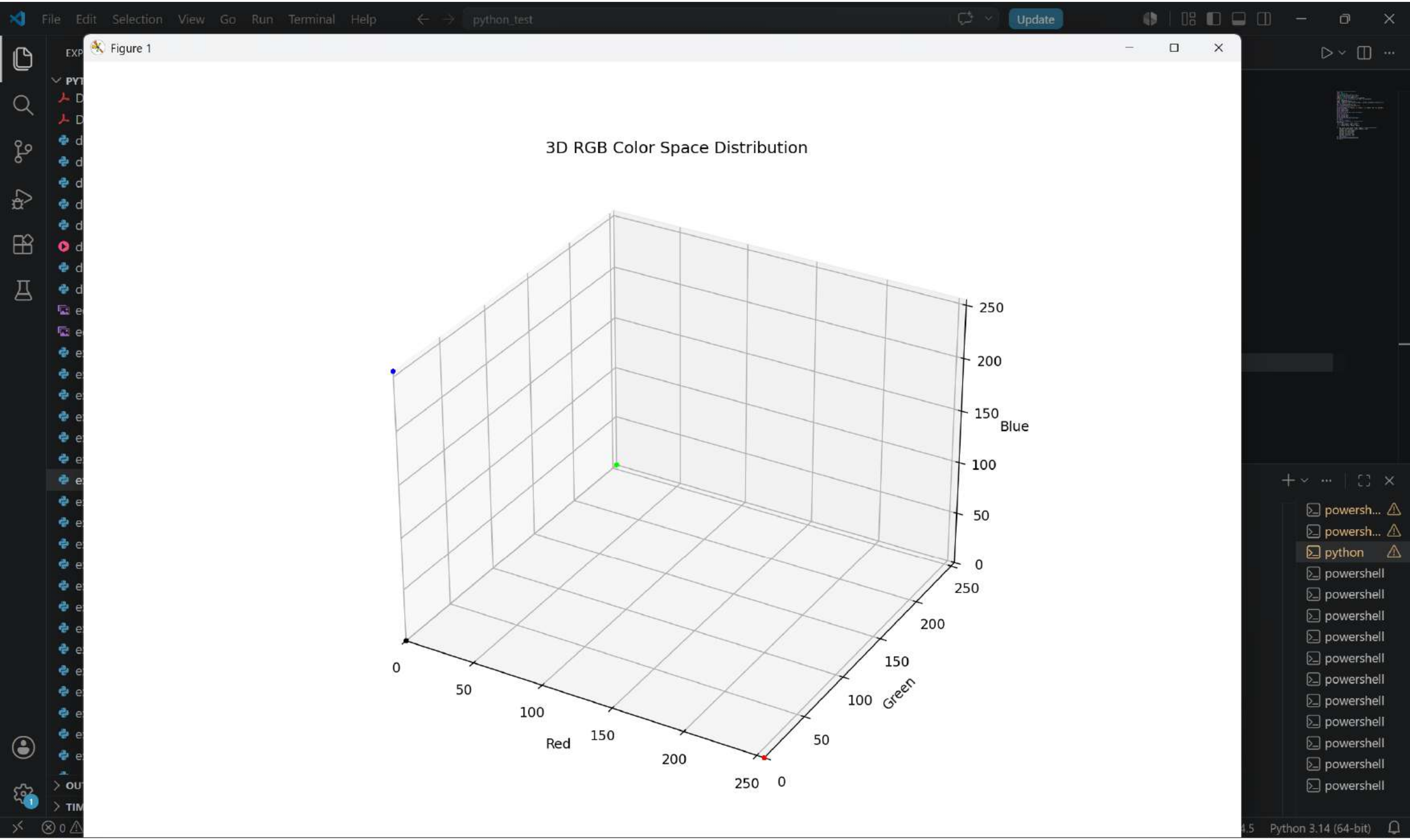
Figure 1
Original
Grayscale
Threshold = 128
Grayscale Histogram
Threshold=128
Count
Intensity
Cumulative Histogram
50%
Cumulative Pixels
Intensity
Statistics
Mean: 27.9
Std Dev: 50.4
Min: 0
Max: 150
Median: 0.0
Threshold: 128
Pixels below: 141844
Pixels above: 18156

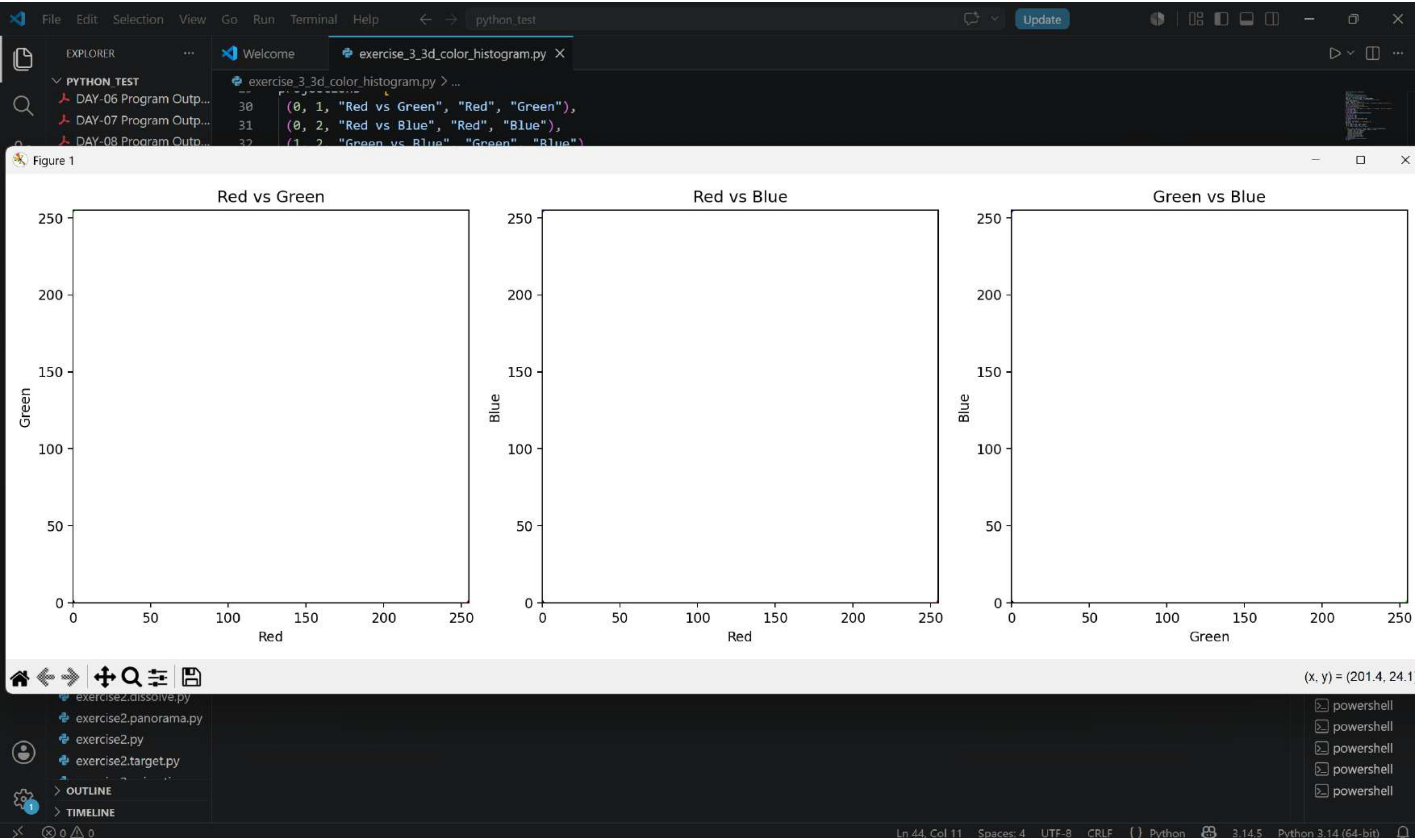

File Edit Selection View Go Run Terminal Help
python_test
Update
EXPLORER
PYTHON_TEST
DAY-08 Program Outp...
day5_transformations....
day6_arithmetic_bitwis...
day7.mini.challenges.py
day8_color_spaces.py
day9_histograms.py
dissolve.mp4
drawing_basics.py
drawing.py
equalized_image.png
exercise_1_channel_art...
exercise_1_custom_pip...
exercise_1_histogram_...
exercise_2_color_picke...
exercise_2_compariso...
exercise_3_color_isolat...
exercise_4_color_spac...
exercise1.brigthness.c...
exercise1.day3.py
exercise1.house.py
exercise1.py
exercise1.thumbnail.py
exercise2.day3.py
exercise2.dissolve.py
exercise2.panorama.py
exercise2.py
exercise2.target.py
exercise3.animation.py
exercise3.chessboard.py
exercise3.day3.py
exercise3.py
exercise3.toolmask.py
OUTLINE
TIMELINE
Welcome
exercise_1_histogram_analyzer.py
exercise_1_histogram_analyzer.py > ...
def analyze_histogram_interactive(image_path):
def update(threshold_val):
stats,
fontsize=10,
verticalalignment='center'
)
plt.tight_layout(rect=[0, 0.08, 1, 1])
plt.draw()
# Create slider
plt.subplots_adjust(bottom=0.15)
ax_slider = plt.axes([0.2, 0.03, 0.6, 0.03])
slider = Slider(
ax=ax_slider,
label='Threshold',
valmin=0,
valmax=255,
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_1_histogram_analyzer.py
PS C:\Users\raghu\OneDrive\Desktop\python_test>
powershell
Ln 159, Col 48 Spaces: 4 UTF-8 CRLF { } Python 3.14.5 Python 3.14 (64-bit)

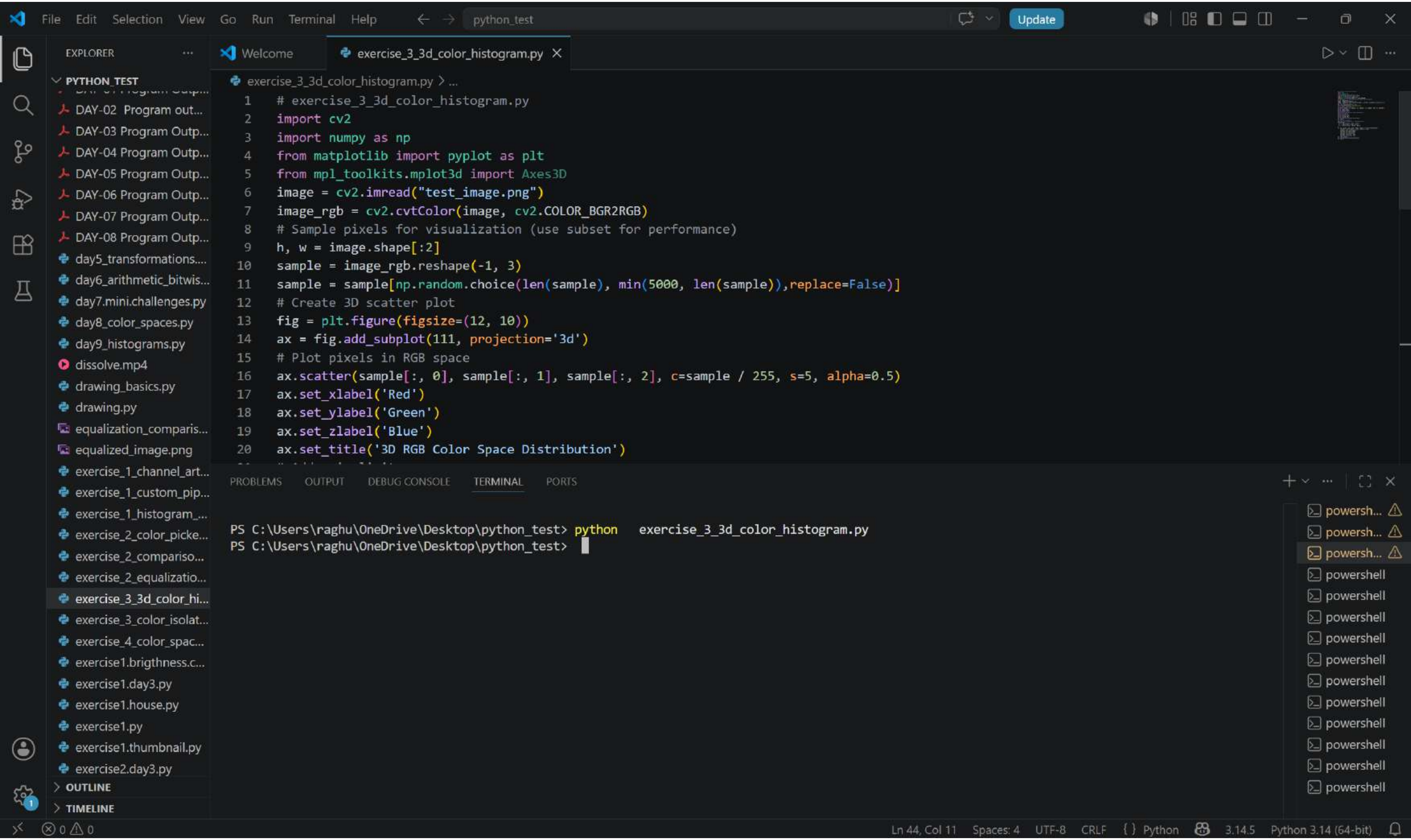


File Edit Selection View Go Run Terminal Help
python_test
Update
EXPLORER
Welcome
exercise_2_equalization_grid.py
Figure 1
Original
Original Histogram
Standard Equalization
Standard Equalization Histogram
CLAHE (Contrast Limiting)
CLAHE (Contrast Limiting) Histogram
Adaptive Histogram
Adaptive Histogram Histogram

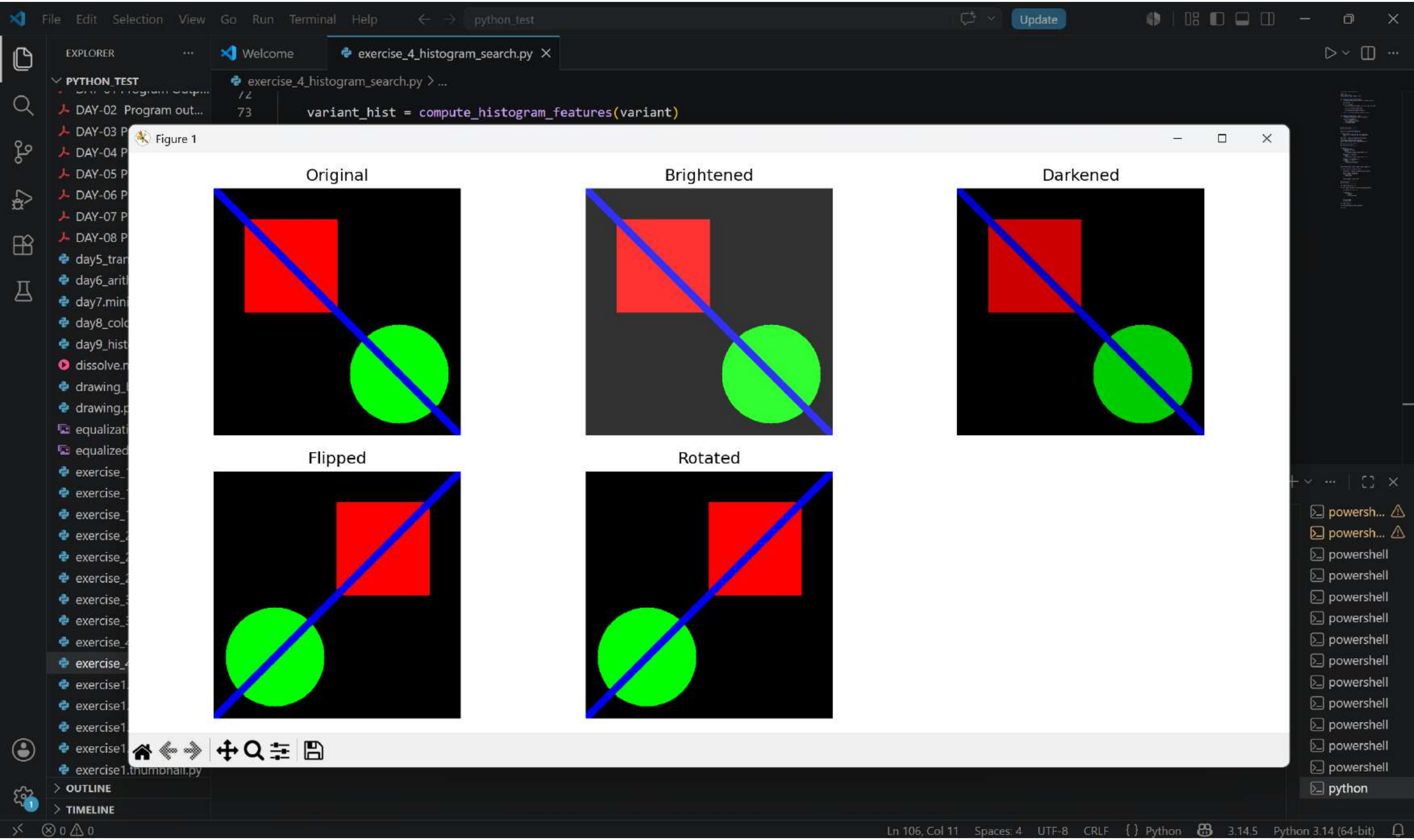

File Edit Selection View Go Run Terminal Help
python_test
Update
EXPLORER
Welcome
exercise_2_equalization_grid.py
PYTHON_TEST
DAY-08 Program Outp...
day5_transformations....
day6_arithmetic_bitwis...
day7.mini.challenges.py
day8_color_spaces.py
day9_histograms.py
dissolve.mp4
drawing_basics.py
drawing.py
equalization_comparis...
equalized_image.png
exercise_1_channel_art...
exercise_1_custom_pip...
exercise_1_histogram_...
exercise_2_color_picke...
exercise_2_compariso...
exercise_2_equalizatio...
exercise_3_color_isolat...
exercise_4_color_spac...
exercise1.brigthness.c...
exercise1.day3.py
exercise1.house.py
exercise1.py
exercise1.thumbnail.py
exercise2.day3.py
exercise2.dissolve.py
exercise2.panorama.py
exercise2.py
exercise2.target.py
exercise3.animation.py
exercise3.chessboard.py
exercise3.day3.py
OUTLINE
TIMELINE
exercise_2_equalization_grid.py > ...
# exercise_2_equalization_grid.py
import cv2
import numpy as np
from matplotlib import pyplot as plt
image = cv2.imread("test_image.png")
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Apply different equalization methods
methods = {
"Original": gray,
"Standard Equalization": cv2.equalizeHist(gray),
"CLAHE (Contrast Limiting)": cv2.createCLAHE(clipLimit=2.0,
tileGridSize=(8,8)).apply(gray),
"Adaptive Histogram": cv2.createCLAHE(clipLimit=3.0,
tileGridSize=(16,16)).apply(gray)
}
# Create comparison grid
fig, axes = plt.subplots(2, 4, figsize=(14, 8))
for idx, (name, img) in enumerate(methods.items()):
row = idx // 2
col = (idx % 2) * 2
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_2_equalization_grid.py
PS C:\Users\raghu\OneDrive\Desktop\python_test>
powersh...
powershell
Ln 35, Col 11 Spaces: 4 UTF-8 CRLF { } Python 3.14.5 Python 3.14 (64-bit)

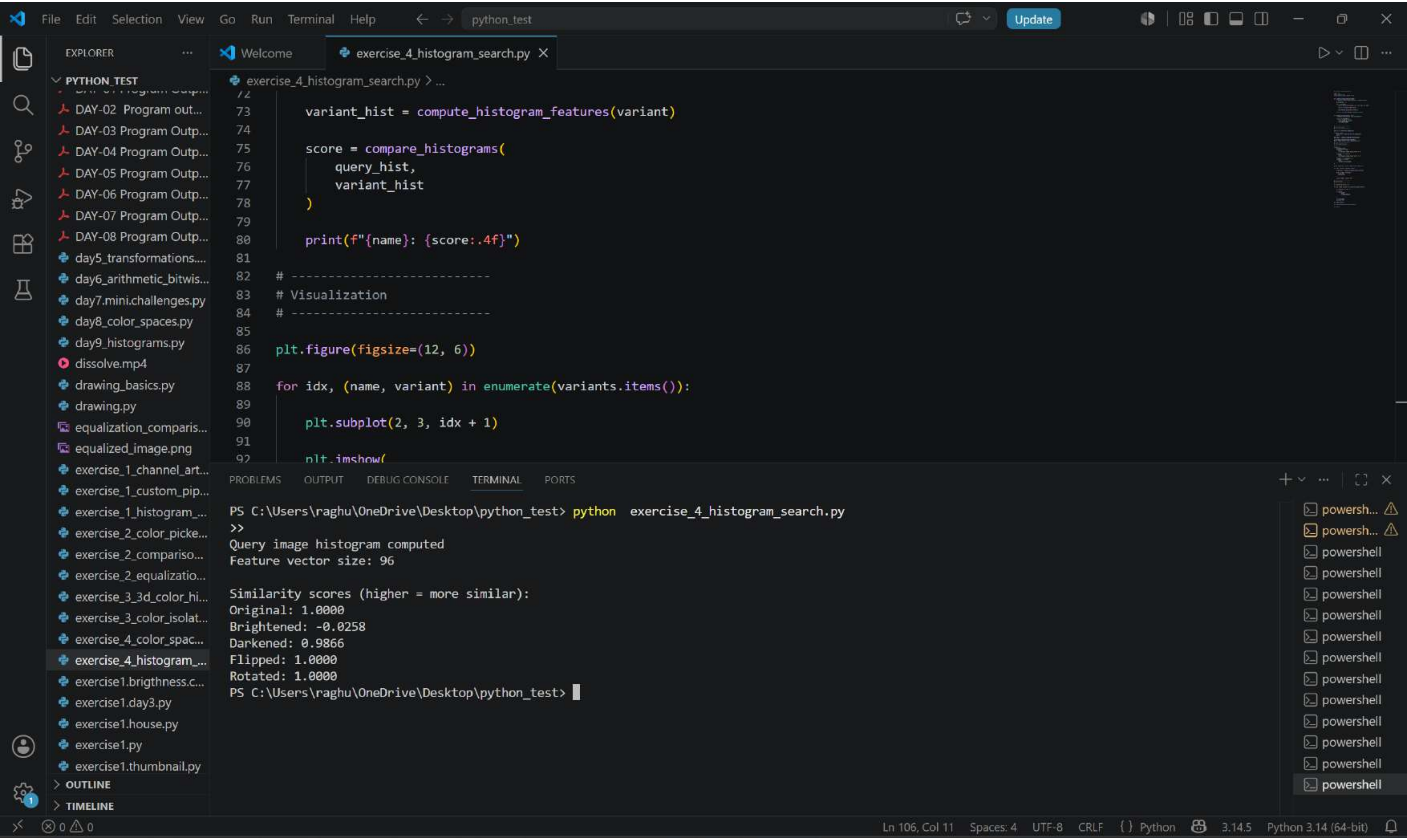

File Edit Selection View Go Run Terminal Help
python_test
Update
Figure 1
3D RGB Color Space Distribution
Red
Green
Blue
0
50
100
150
200
250
powershell
python
Python 3.14 (64-bit)


File Edit Selection View Go Run Terminal Help
python_test
Update
EXPLORER
PYTHON_TEST
DAY-06 Program Outp...
DAY-07 Program Outp...
DAY-08 Program Outp...
Welcome
exercise_3_3d_color_histogram.py
(0, 1, "Red vs Green", "Red", "Green"),
(0, 2, "Red vs Blue", "Red", "Blue"),
Figure 1
Red vs Green
Red vs Blue
Green vs Blue
Green
Blue
Red
(x, y) = (201.4, 24.1
exercise2.panorama.py
exercise2.py
exercise2.target.py
OUTLINE
TIMELINE
powershell
Ln 44, Col 11 Spaces: 4 UTF-8 CRLF Python 3.14.5 Python 3.14 (64-bit)



python_test
Welcome
exercise_3_3d_color_histogram.py
# exercise_3_3d_color_histogram.py
import cv2
import numpy as np
from matplotlib import pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
image = cv2.imread("test_image.png")
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
# Sample pixels for visualization (use subset for performance)
h, w = image.shape[:2]
sample = image_rgb.reshape(-1, 3)
sample = sample[np.random.choice(len(sample), min(5000, len(sample)),replace=False)]
# Create 3D scatter plot
fig = plt.figure(figsize=(12, 10))
ax = fig.add_subplot(111, projection='3d')
# Plot pixels in RGB space
ax.scatter(sample[:, 0], sample[:, 1], sample[:, 2], c=sample / 255, s=5, alpha=0.5)
ax.set_xlabel('Red')
ax.set_ylabel('Green')
ax.set_zlabel('Blue')
ax.set_title('3D RGB Color Space Distribution')
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_3_3d_color_histogram.py
PS C:\Users\raghu\OneDrive\Desktop\python_test>
Ln 44, Col 11 Spaces: 4 UTF-8 CRLF Python 3.14.5 Python 3.14 (64-bit)


File Edit Selection View Go Run Terminal Help
python_test
Update
EXPLORER
PYTHON_TEST
DAY-02 Program out...
exercise_4_histogram_search.py
variant_hist = compute_histogram_features(variant)
Figure 1
Original
Brightened
Darkened
Flipped
Rotated
OUTLINE
TIMELINE
powershell
python
Ln 106, Col 11 Spaces: 4 UTF-8 CRLF { } Python 3.14.5 Python 3.14 (64-bit)


exercise_4_histogram_search.py
variant_hist = compute_histogram_features(variant)
score = compare_histograms(
query_hist,
variant_hist
)
print(f"{name}: {score:.4f}")
# --------------------------
# Visualization
# --------------------------
plt.figure(figsize=(12, 6))
for idx, (name, variant) in enumerate(variants.items()):
plt.subplot(2, 3, idx + 1)
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_4_histogram_search.py
>>
Query image histogram computed
Feature vector size: 96
Similarity scores (higher = more similar):
Original: 1.0000
Brightened: -0.0258
Darkened: 0.9866
Flipped: 1.0000
Rotated: 1.0000
PS C:\Users\raghu\OneDrive\Desktop\python_test>

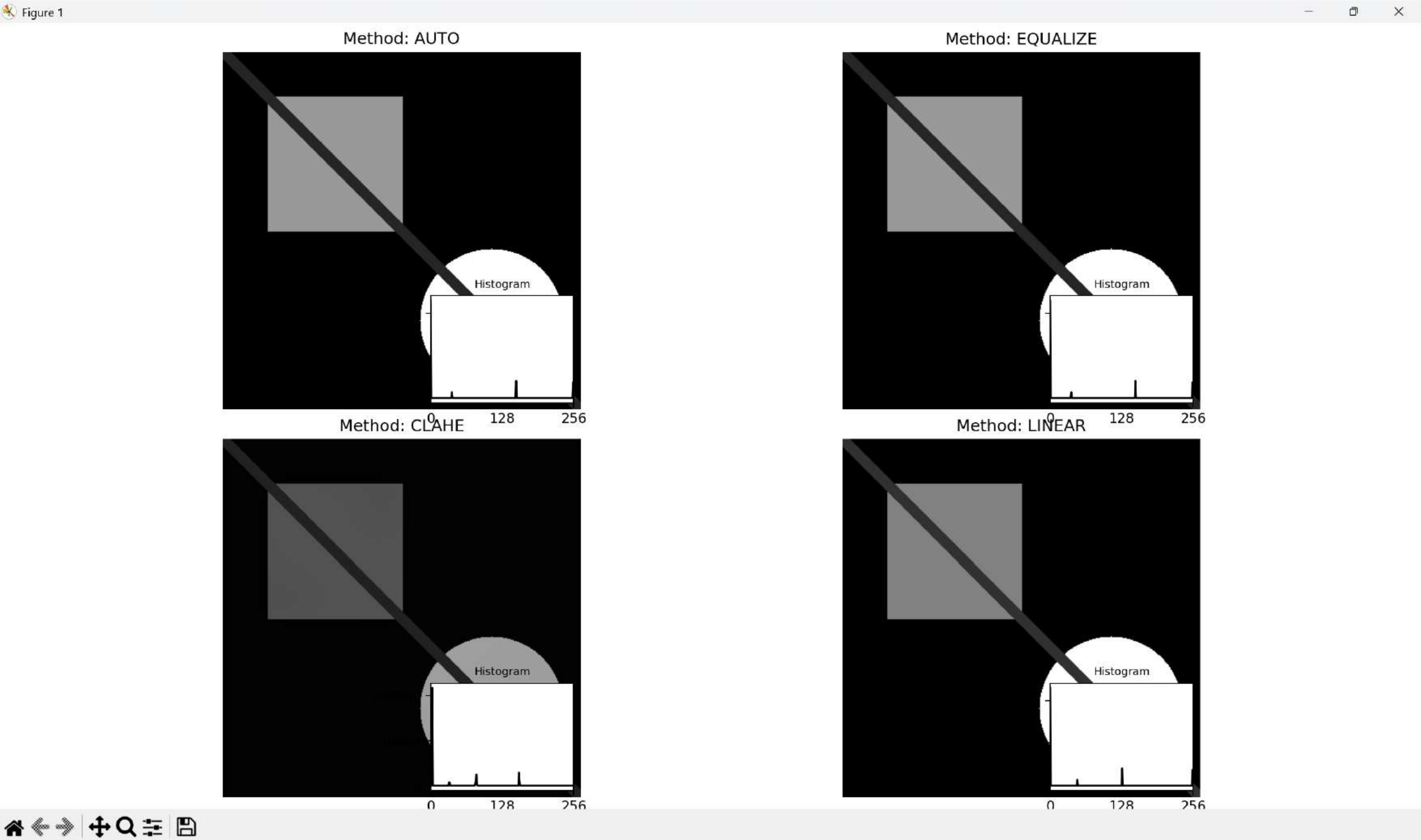
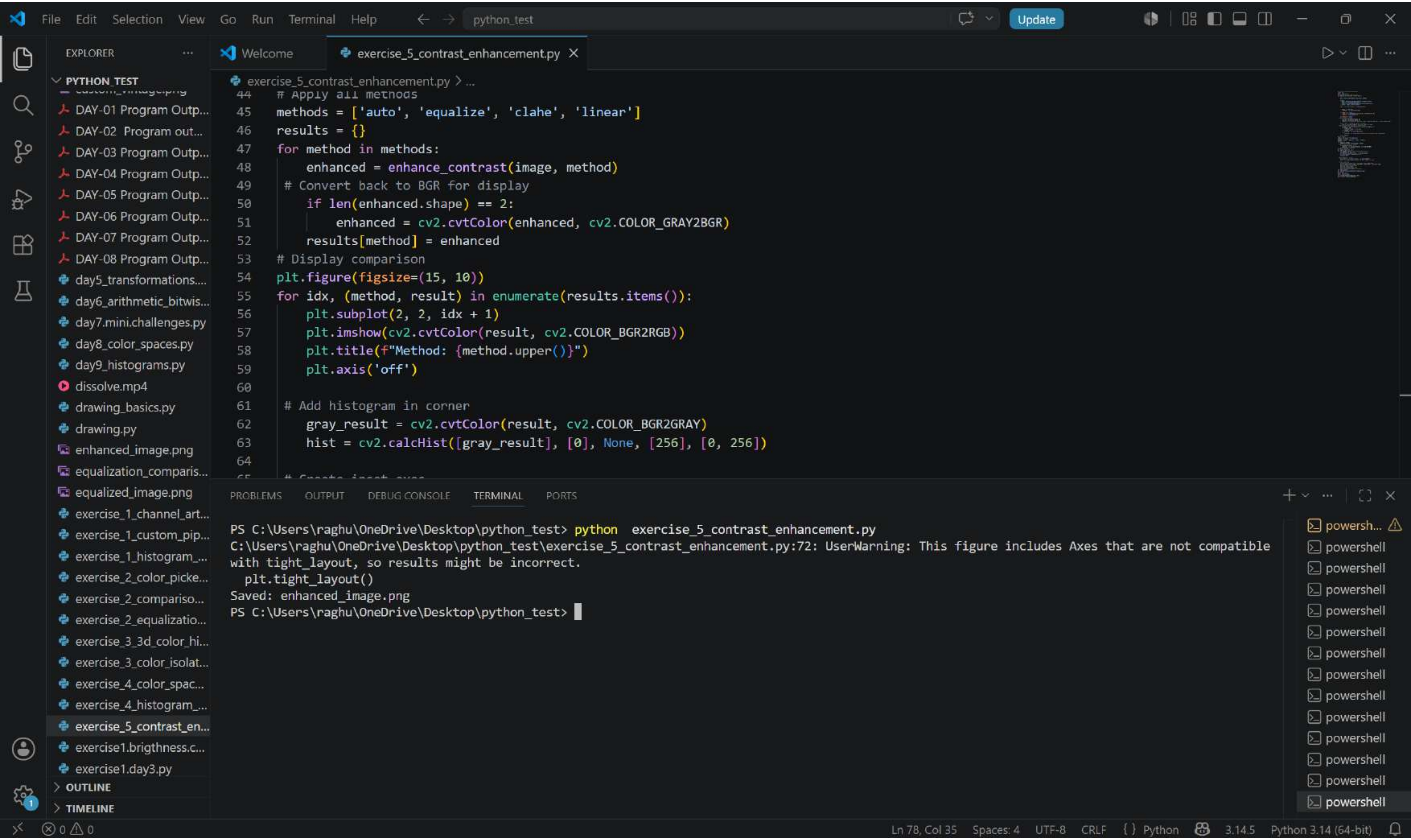
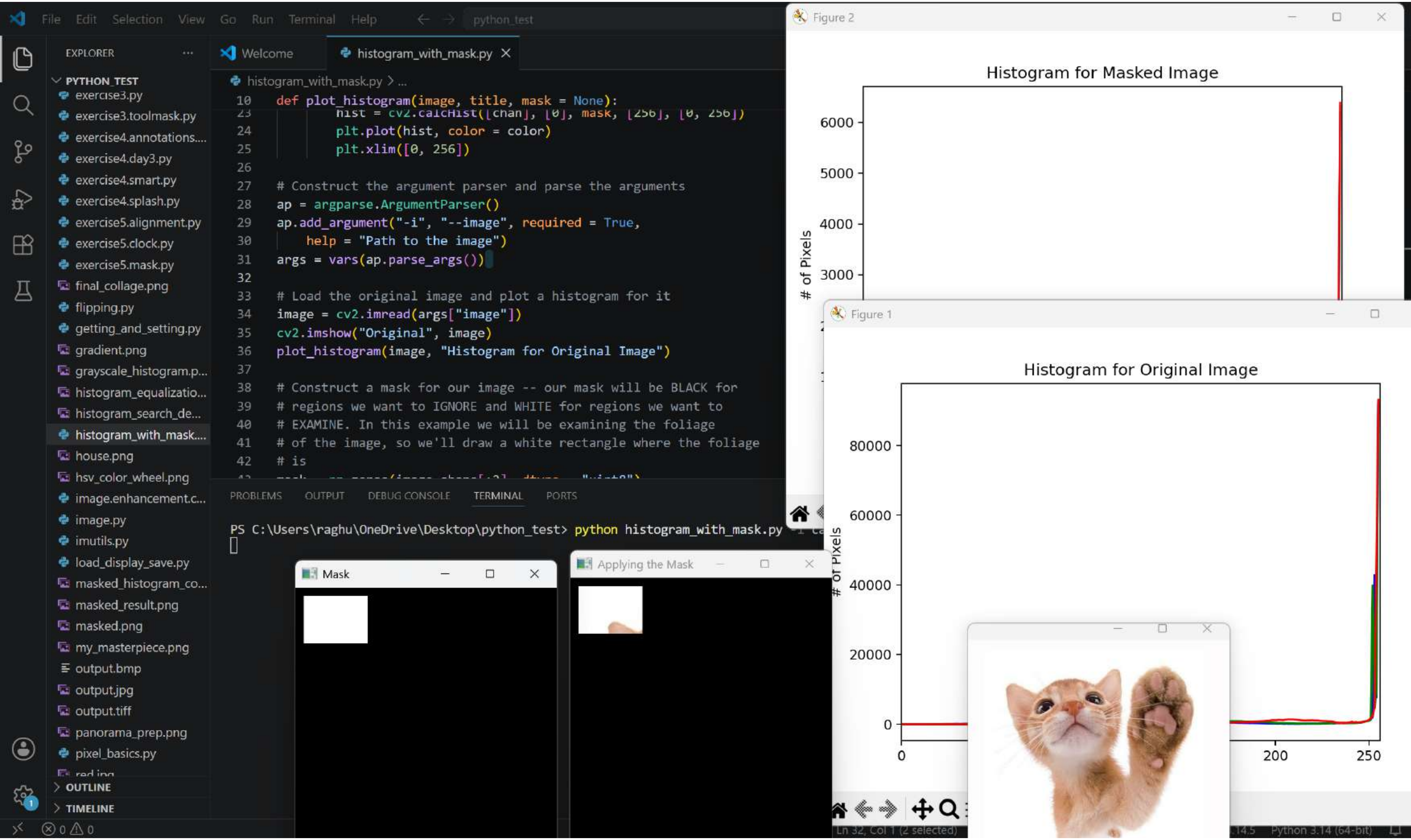
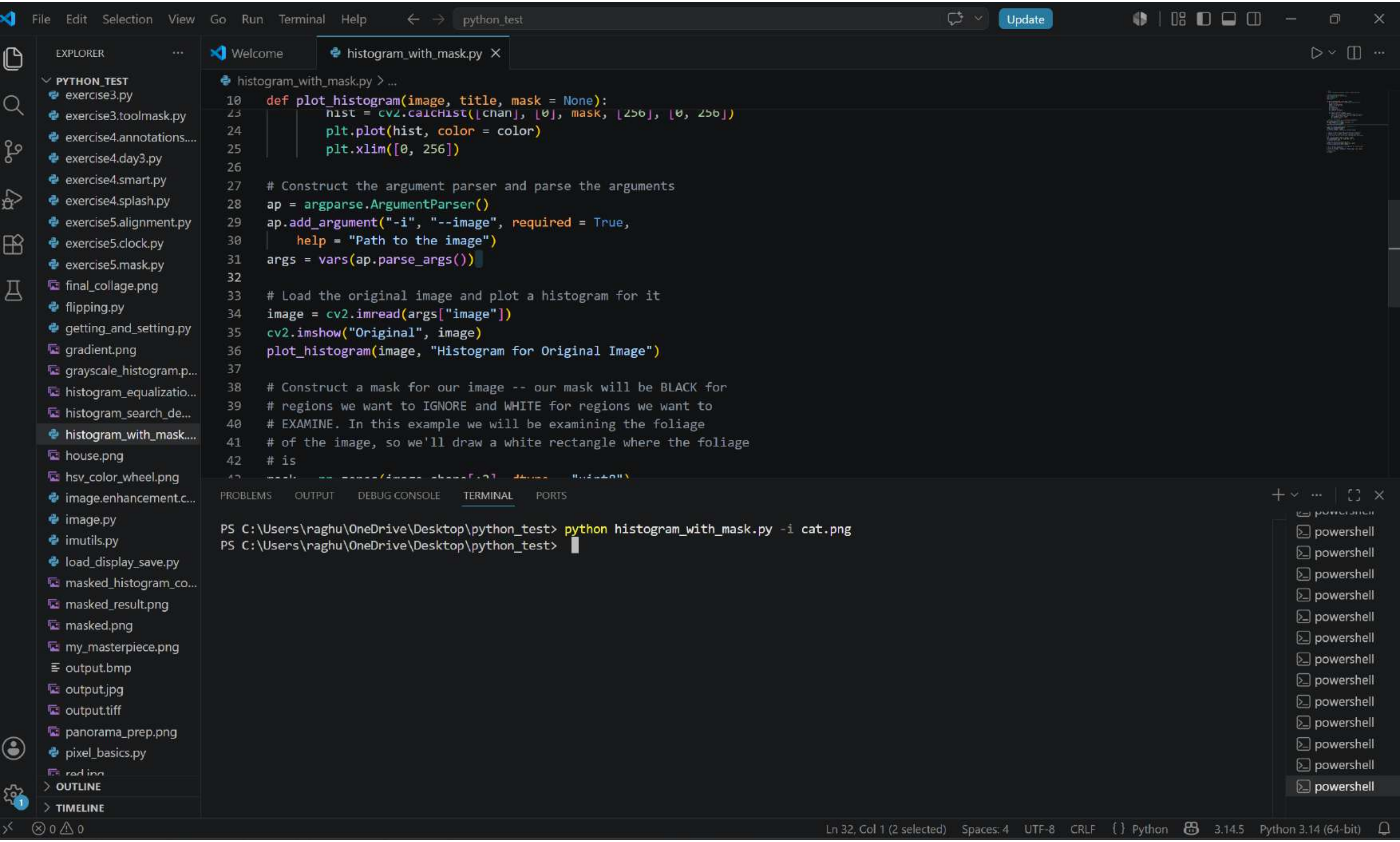


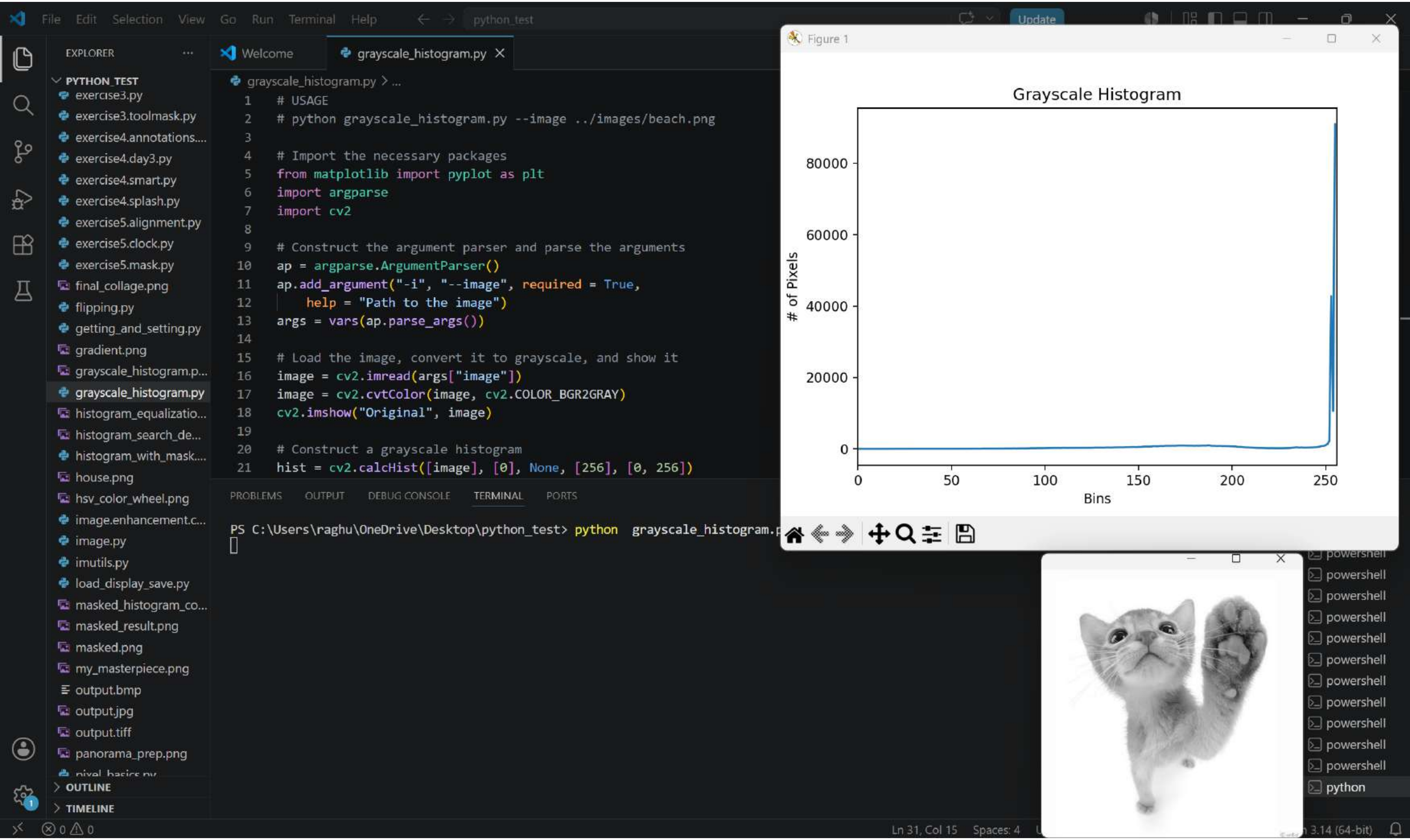
Figure 1
Method: AUTO
Method: EQUALIZE
Method: CLAHE
Method: LINEAR
Histogram
0
128
256

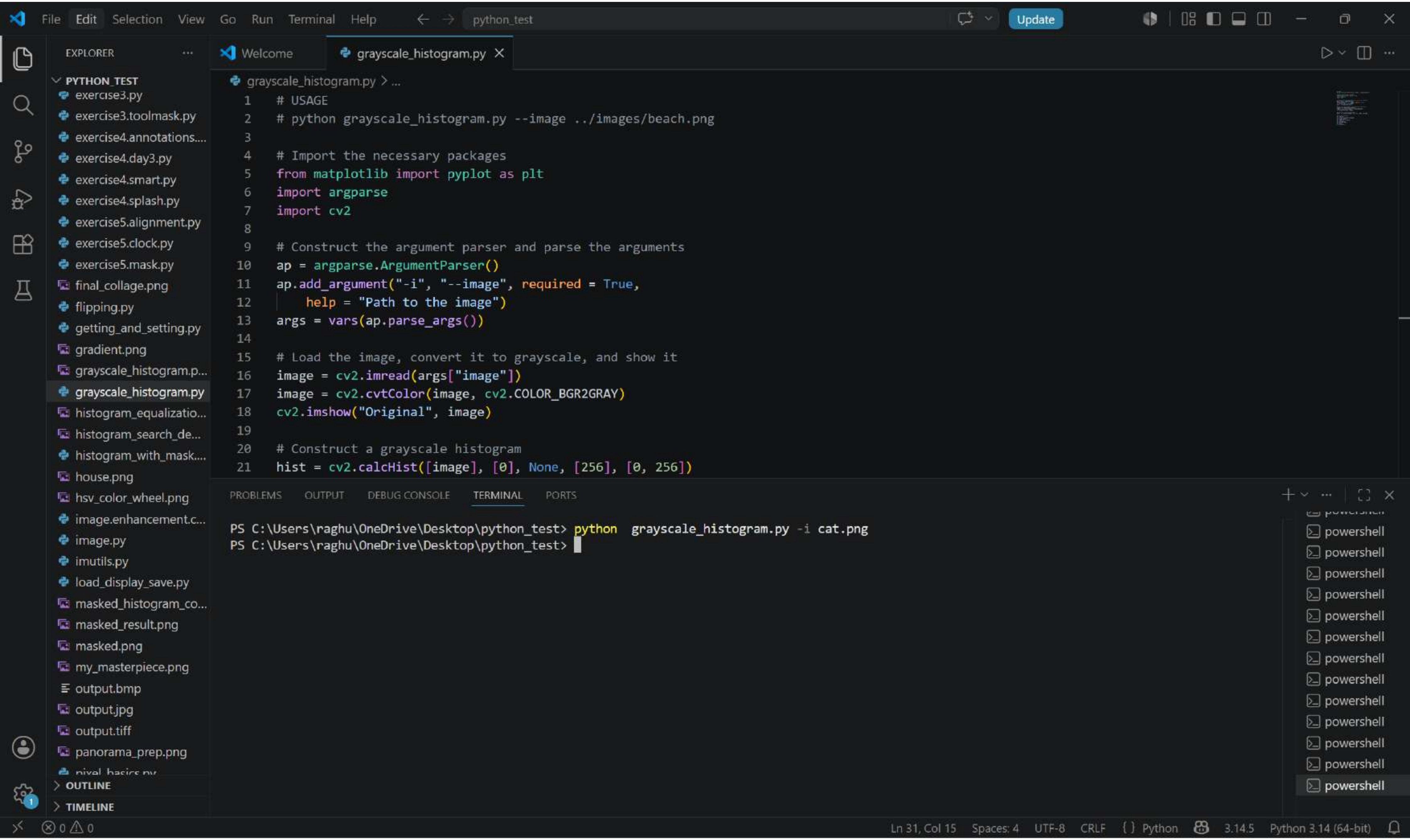


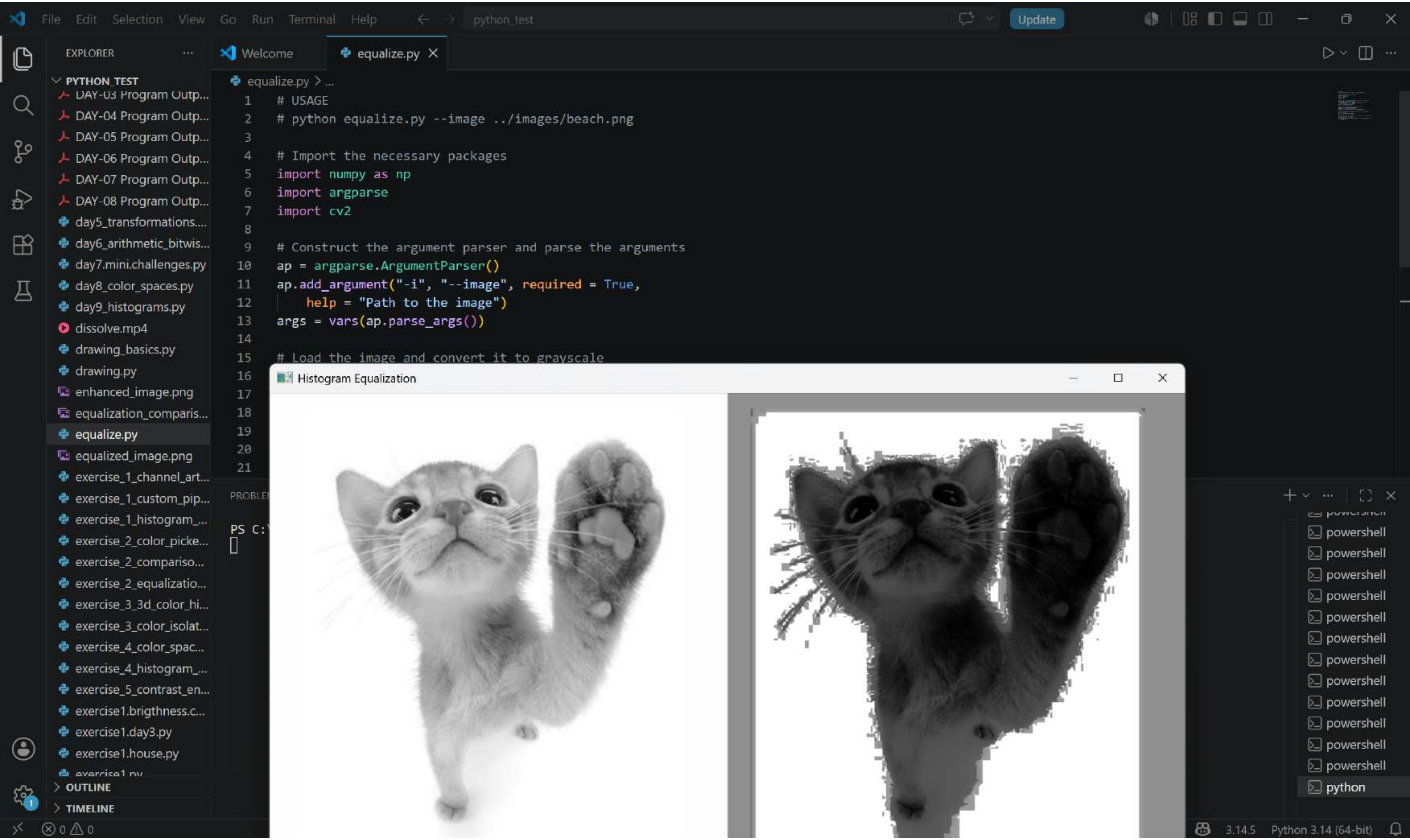
File Edit Selection View Go Run Terminal Help
python_test
Update
EXPLORER
PYTHON_TEST
DAY-01 Program Outp...
DAY-02 Program out...
DAY-03 Program Outp...
DAY-04 Program Outp...
DAY-05 Program Outp...
DAY-06 Program Outp...
DAY-07 Program Outp...
DAY-08 Program Outp...
day5_transformations....
day6_arithmetic_bitwis...
day7.mini.challenges.py
day8_color_spaces.py
day9_histograms.py
dissolve.mp4
drawing_basics.py
drawing.py
enhanced_image.png
equalization_comparis...
equalized_image.png
exercise_1_channel_art...
exercise_1_custom_pip...
exercise_1_histogram_...
exercise_2_color_picke...
exercise_2_compariso...
exercise_2_equalizatio...
exercise_3_3d_color_hi...
exercise_3_color_isolat...
exercise_4_color_spac...
exercise_4_histogram_...
exercise_5_contrast_en...
exercise1.brigthness.c...
exercise1.day3.py
OUTLINE
TIMELINE
Welcome
exercise_5_contrast_enhancement.py
exercise_5_contrast_enhancement.py > ...
methods = ['auto', 'equalize', 'clahe', 'linear']
results = {}
for method in methods:
    enhanced = enhance_contrast(image, method)
 # Convert back to BGR for display
    if len(enhanced.shape) == 2:
        enhanced = cv2.cvtColor(enhanced, cv2.COLOR_GRAY2BGR)
    results[method] = enhanced
# Display comparison
plt.figure(figsize=(15, 10))
for idx, (method, result) in enumerate(results.items()):
    plt.subplot(2, 2, idx + 1)
    plt.imshow(cv2.cvtColor(result, cv2.COLOR_BGR2RGB))
    plt.title(f"Method: {method.upper()}")
    plt.axis('off')
 # Add histogram in corner
    gray_result = cv2.cvtColor(result, cv2.COLOR_BGR2GRAY)
    hist = cv2.calcHist([gray_result], [0], None, [256], [0, 256])
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_5_contrast_enhancement.py
C:\Users\raghu\OneDrive\Desktop\python_test\exercise_5_contrast_enhancement.py:72: UserWarning: This figure includes Axes that are not compatible with tight_layout, so results might be incorrect.
  plt.tight_layout()
Saved: enhanced_image.png
PS C:\Users\raghu\OneDrive\Desktop\python_test>
powershell
Ln 78, Col 35 Spaces: 4 UTF-8 CRLF { } Python 3.14.5 Python 3.14 (64-bit)

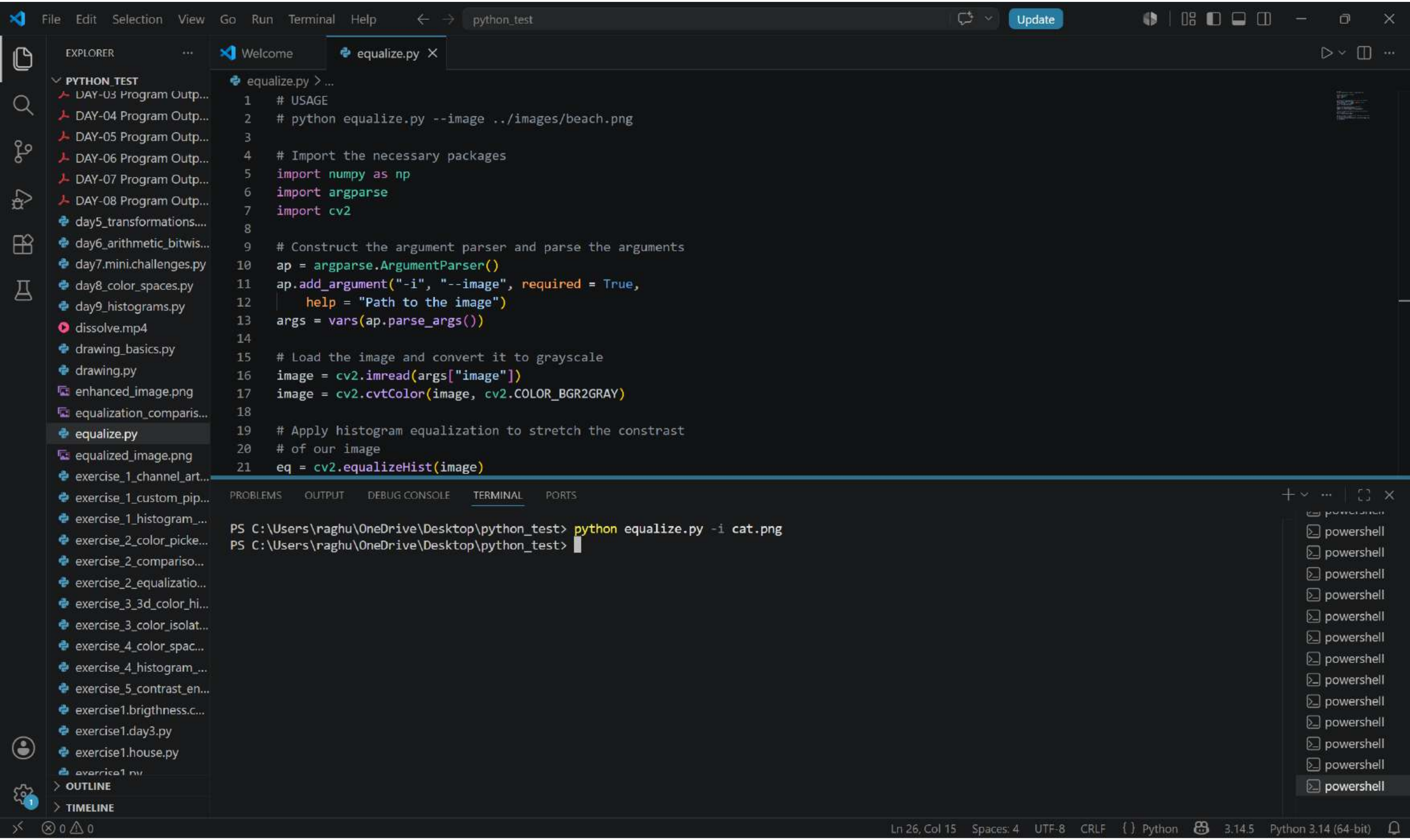

File Edit Selection View Go Run Terminal Help
python_test
EXPLORER
PYTHON_TEST
exercise3.py
exercise3.toolmask.py
exercise4.annotations....
exercise4.day3.py
exercise4.smart.py
exercise4.splash.py
exercise5.alignment.py
exercise5.clock.py
exercise5.mask.py
final_collage.png
flipping.py
getting_and_setting.py
gradient.png
grayscale_histogram.p...
histogram_equalizatio...
histogram_search_de...
histogram_with_mask....
house.png
hsv_color_wheel.png
image.enhancement.c...
image.py
imutils.py
load_display_save.py
masked_histogram_co...
masked_result.png
masked.png
my_masterpiece.png
output.bmp
output.jpg
output.tiff
panorama_prep.png
pixel_basics.py
OUTLINE
TIMELINE
Welcome
histogram_with_mask.py
histogram_with_mask.py > ...
10 def plot_histogram(image, title, mask = None):
24 plt.plot(hist, color = color)
25 plt.xlim([0, 256])
27 # Construct the argument parser and parse the arguments
28 ap = argparse.ArgumentParser()
29 ap.add_argument("-i", "--image", required = True,
30 help = "Path to the image")
31 args = vars(ap.parse_args())
33 # Load the original image and plot a histogram for it
34 image = cv2.imread(args["image"])
35 cv2.imshow("Original", image)
36 plot_histogram(image, "Histogram for Original Image")
38 # Construct a mask for our image -- our mask will be BLACK for
39 # regions we want to IGNORE and WHITE for regions we want to
40 # EXAMINE. In this example we will be examining the foliage
41 # of the image, so we'll draw a white rectangle where the foliage
42 # is
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python histogram_with_mask.py
Mask
Applying the Mask
Figure 2
Histogram for Masked Image
# of Pixels
6000
5000
4000
3000
Figure 1
Histogram for Original Image
# of Pixels
80000
60000
40000
20000
0
0
200
250
Ln 32, Col 1 (2 selected)
Python 3.14 (64-bit)


File Edit Selection View Go Run Terminal Help
python_test
Update
EXPLORER
PYTHON_TEST
exercise3.py
exercise3.toolmask.py
exercise4.annotations....
exercise4.day3.py
exercise4.smart.py
exercise4.splash.py
exercise5.alignment.py
exercise5.clock.py
exercise5.mask.py
final_collage.png
flipping.py
getting_and_setting.py
gradient.png
grayscale_histogram.p...
histogram_equalizatio...
histogram_search_de...
histogram_with_mask....
house.png
hsv_color_wheel.png
image.enhancement.c...
image.py
imutils.py
load_display_save.py
masked_histogram_co...
masked_result.png
masked.png
my_masterpiece.png
output.bmp
output.jpg
output.tiff
panorama_prep.png
pixel_basics.py
OUTLINE
TIMELINE
Welcome
histogram_with_mask.py
histogram_with_mask.py > ...
def plot_histogram(image, title, mask = None):
plt.plot(hist, color = color)
plt.xlim([0, 256])
# Construct the argument parser and parse the arguments
ap = argparse.ArgumentParser()
ap.add_argument("-i", "--image", required = True,
help = "Path to the image")
args = vars(ap.parse_args())
# Load the original image and plot a histogram for it
image = cv2.imread(args["image"])
cv2.imshow("Original", image)
plot_histogram(image, "Histogram for Original Image")
# Construct a mask for our image -- our mask will be BLACK for
# regions we want to IGNORE and WHITE for regions we want to
# EXAMINE. In this example we will be examining the foliage
# of the image, so we'll draw a white rectangle where the foliage
# is
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python histogram_with_mask.py -i cat.png
PS C:\Users\raghu\OneDrive\Desktop\python_test>
powershell
Ln 32, Col 1 (2 selected) Spaces: 4 UTF-8 CRLF { } Python 3.14.5 Python 3.14 (64-bit)


File Edit Selection View Go Run Terminal Help
python_test
Update
EXPLORER
PYTHON_TEST
exercise3.py
exercise3.toolmask.py
exercise4.annotations....
exercise4.day3.py
exercise4.smart.py
exercise4.splash.py
exercise5.alignment.py
exercise5.clock.py
exercise5.mask.py
final_collage.png
flipping.py
getting_and_setting.py
gradient.png
grayscale_histogram.p...
grayscale_histogram.py
histogram_equalizatio...
histogram_search_de...
histogram_with_mask....
house.png
hsv_color_wheel.png
image.enhancement.c...
image.py
imutils.py
load_display_save.py
masked_histogram_co...
masked_result.png
masked.png
my_masterpiece.png
output.bmp
output.jpg
output.tiff
panorama_prep.png
OUTLINE
TIMELINE
Welcome
grayscale_histogram.py
grayscale_histogram.py > ...
1 # USAGE
2 # python grayscale_histogram.py --image ../images/beach.png
3
4 # Import the necessary packages
5 from matplotlib import pyplot as plt
6 import argparse
7 import cv2
8
9 # Construct the argument parser and parse the arguments
10 ap = argparse.ArgumentParser()
11 ap.add_argument("-i", "--image", required = True,
12 help = "Path to the image")
13 args = vars(ap.parse_args())
14
15 # Load the image, convert it to grayscale, and show it
16 image = cv2.imread(args["image"])
17 image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
18 cv2.imshow("Original", image)
19
20 # Construct a grayscale histogram
21 hist = cv2.calcHist([image], [0], None, [256], [0, 256])
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python grayscale_histogram.p
Figure 1
Grayscale Histogram
# of Pixels
80000
60000
40000
20000
0
0 50 100 150 200 250
Bins
powershell
powershell
powershell
powershell
powershell
powershell
powershell
powershell
powershell
powershell
powershell
python
Ln 31, Col 15 Spaces: 4
3.14 (64-bit)


File Edit Selection View Go Run Terminal Help
python_test
Update
EXPLORER
PYTHON_TEST
exercise3.py
exercise3.toolmask.py
exercise4.annotations....
exercise4.day3.py
exercise4.smart.py
exercise4.splash.py
exercise5.alignment.py
exercise5.clock.py
exercise5.mask.py
final_collage.png
flipping.py
getting_and_setting.py
gradient.png
grayscale_histogram.p...
grayscale_histogram.py
histogram_equalizatio...
histogram_search_de...
histogram_with_mask....
house.png
hsv_color_wheel.png
image.enhancement.c...
image.py
imutils.py
load_display_save.py
masked_histogram_co...
masked_result.png
masked.png
my_masterpiece.png
output.bmp
output.jpg
output.tiff
panorama_prep.png
OUTLINE
TIMELINE
Welcome
grayscale_histogram.py
grayscale_histogram.py > ...
# USAGE
# python grayscale_histogram.py --image ../images/beach.png
# Import the necessary packages
from matplotlib import pyplot as plt
import argparse
import cv2
# Construct the argument parser and parse the arguments
ap = argparse.ArgumentParser()
ap.add_argument("-i", "--image", required = True,
help = "Path to the image")
args = vars(ap.parse_args())
# Load the image, convert it to grayscale, and show it
image = cv2.imread(args["image"])
image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
cv2.imshow("Original", image)
# Construct a grayscale histogram
hist = cv2.calcHist([image], [0], None, [256], [0, 256])
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python grayscale_histogram.py -i cat.png
PS C:\Users\raghu\OneDrive\Desktop\python_test>
powershell
Ln 31, Col 15 Spaces: 4 UTF-8 CRLF { } Python 3.14.5 Python 3.14 (64-bit)


File Edit Selection View Go Run Terminal Help
python_test
Update
EXPLORER
PYTHON_TEST
DAY-03 Program Outp...
DAY-04 Program Outp...
DAY-05 Program Outp...
DAY-06 Program Outp...
DAY-07 Program Outp...
DAY-08 Program Outp...
day5_transformations....
day6_arithmetic_bitwis...
day7.mini.challenges.py
day8_color_spaces.py
day9_histograms.py
dissolve.mp4
drawing_basics.py
drawing.py
enhanced_image.png
equalization_comparis...
equalize.py
equalized_image.png
exercise_1_channel_art...
exercise_1_custom_pip...
exercise_1_histogram_...
exercise_2_color_picke...
exercise_2_compariso...
exercise_2_equalizatio...
exercise_3_3d_color_hi...
exercise_3_color_isolat...
exercise_4_color_spac...
exercise_4_histogram_...
exercise_5_contrast_en...
exercise1.brigthness.c...
exercise1.day3.py
exercise1.house.py
OUTLINE
TIMELINE
Welcome
equalize.py
equalize.py > ...
# USAGE
# python equalize.py --image ../images/beach.png
# Import the necessary packages
import numpy as np
import argparse
import cv2
# Construct the argument parser and parse the arguments
ap = argparse.ArgumentParser()
ap.add_argument("-i", "--image", required = True,
help = "Path to the image")
args = vars(ap.parse_args())
# Load the image and convert it to grayscale
Histogram Equalization
PS C:\
powershell
python
3.14.5
Python 3.14 (64-bit)


File Edit Selection View Go Run Terminal Help
python_test
Update
EXPLORER
PYTHON_TEST
DAY-04 Program Outp...
DAY-05 Program Outp...
DAY-06 Program Outp...
DAY-07 Program Outp...
DAY-08 Program Outp...
day5_transformations....
day6_arithmetic_bitwis...
day7.mini.challenges.py
day8_color_spaces.py
day9_histograms.py
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enhanced_image.png
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equalized_image.png
exercise_1_channel_art...
exercise_1_custom_pip...
exercise_1_histogram_...
exercise_2_color_picke...
exercise_2_compariso...
exercise_2_equalizatio...
exercise_3_3d_color_hi...
exercise_3_color_isolat...
exercise_4_color_spac...
exercise_4_histogram_...
exercise_5_contrast_en...
exercise1.brigthness.c...
exercise1.day3.py
exercise1.house.py
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import cv2
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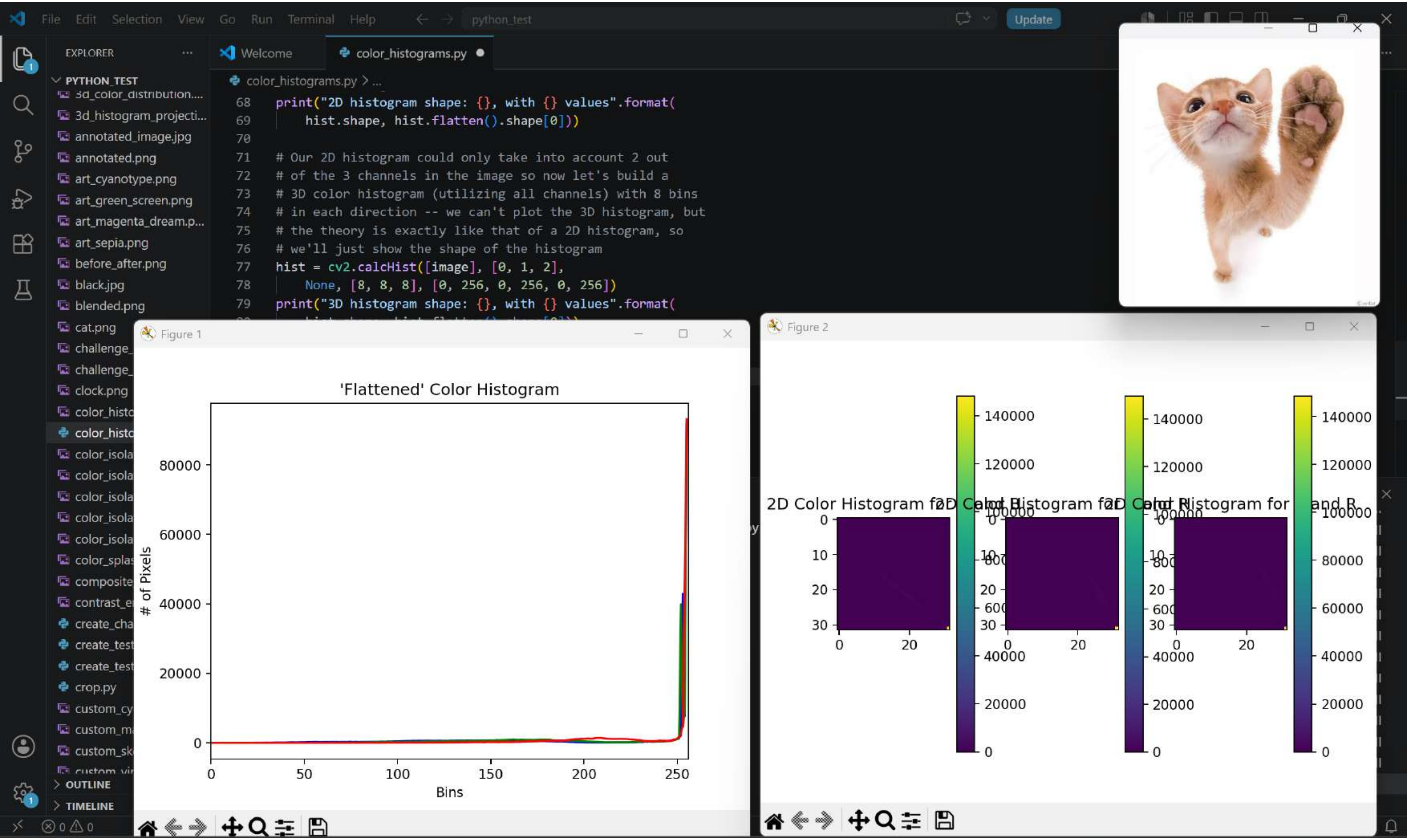
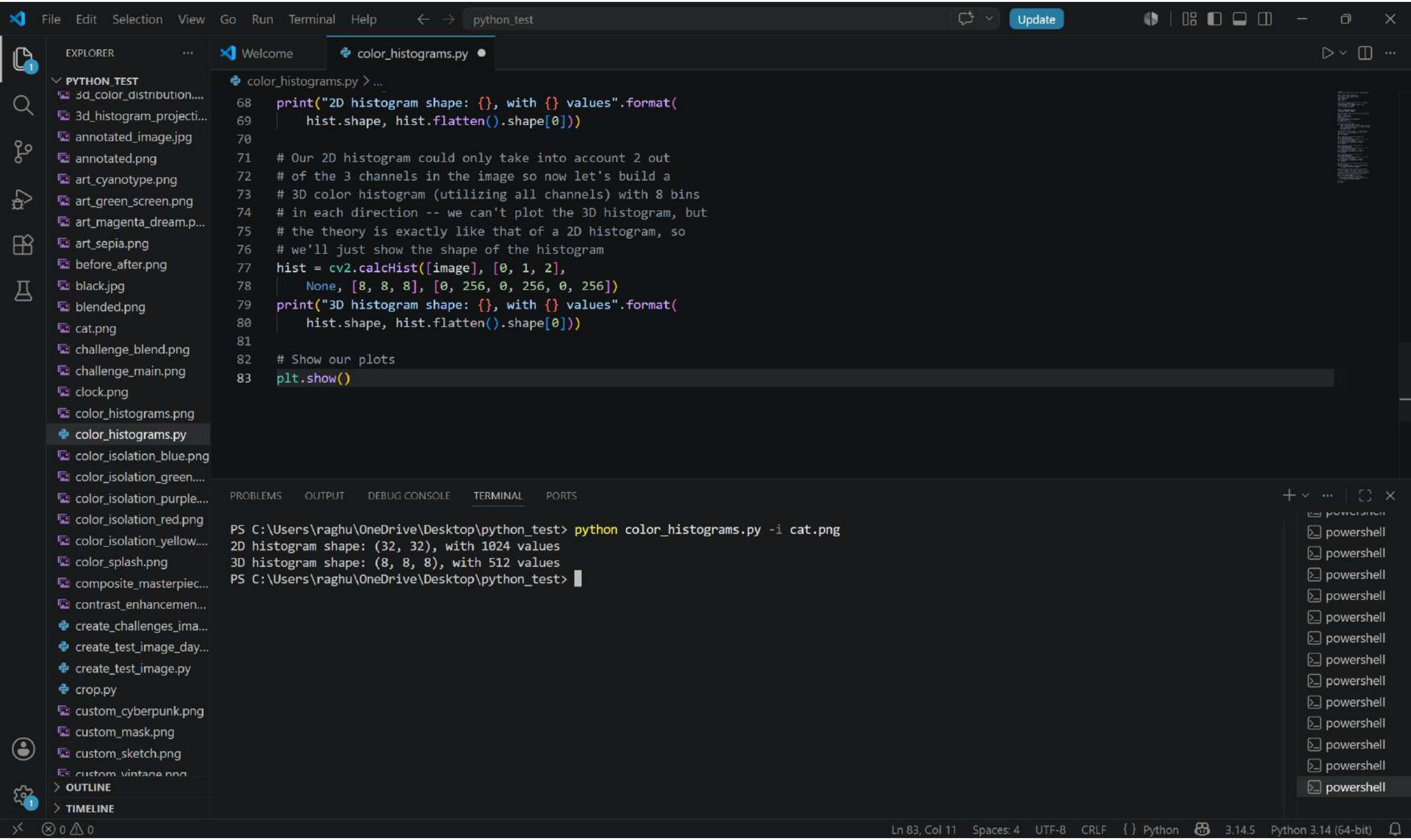
help = "Path to the image")
args = vars(ap.parse_args())
# Load the image and convert it to grayscale
image = cv2.imread(args["image"])
image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Apply histogram equalization to stretch the contrast
# of our image
eq = cv2.equalizeHist(image)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python equalize.py -i cat.png
PS C:\Users\raghu\OneDrive\Desktop\python_test>
powershell
Ln 26, Col 15 Spaces: 4 UTF-8 CRLF { } Python 3.14.5 Python 3.14 (64-bit)


File Edit Selection View Go Run Terminal Help
python_test
Update
EXPLORER
PYTHON_TEST
Welcome
color_histograms.py
color_histograms.py > ...
print("2D histogram shape: {}, with {} values".format(
hist.shape, hist.flatten().shape[0]))
# Our 2D histogram could only take into account 2 out
# of the 3 channels in the image so now let's build a
# 3D color histogram (utilizing all channels) with 8 bins
# in each direction -- we can't plot the 3D histogram, but
# the theory is exactly like that of a 2D histogram, so
# we'll just show the shape of the histogram
hist = cv2.calcHist([image], [0, 1, 2],
None, [8, 8, 8], [0, 256, 0, 256, 0, 256])
print("3D histogram shape: {}, with {} values".format(
3d_histogram_projecti...
annotated_image.jpg
annotated.png
art_cyanotype.png
art_green_screen.png
art_magenta_dream.p...
art_sepia.png
before_after.png
black.jpg
blended.png
cat.png
clock.png
crop.py
OUTLINE
TIMELINE
Figure 1
'Flattened' Color Histogram
# of Pixels
Bins
Figure 2
2D Color Histogram for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EXPLORER
PYTHON_TEST
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# 3D color histogram (utilizing all channels) with 8 bins
# in each direction -- we can't plot the 3D histogram, but
# the theory is exactly like that of a 2D histogram, so
# we'll just show the shape of the histogram
hist = cv2.calcHist([image], [0, 1, 2],
None, [8, 8, 8], [0, 256, 0, 256, 0, 256])
print("3D histogram shape: {}, with {} values".format(
hist.shape, hist.flatten().shape[0]))
# Show our plots
plt.show()
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\raghu\OneDrive\Desktop\python_test> python color_histograms.py -i cat.png
2D histogram shape: (32, 32), with 1024 values
3D histogram shape: (8, 8, 8), with 512 values
PS C:\Users\raghu\OneDrive\Desktop\python_test>
powershell
OUTLINE
TIMELINE
Ln 83, Col 11 Spaces: 4 UTF-8 CRLF Python 3.14.5 Python 3.14 (64-bit)